MILITARY SPECIFICATION

CASE SECTION, BALLISTIC, NOSE

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of ballistic nose section entitled Case Section Ballistic Nose which is used in the Honest John warhead section

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-G-2550	- General Specification for Ammunition Except Small Arms Ammunition.
MIL-T-5021	- Tests; Aircraft Welding Operators Certification.
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.

FSC 1115

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DRAWINGS

ORDNANCE CORPS

8808362	- Container, Packing, Ammunition, For Domestic (Conus) Shipments of Special Weapon Items, Packing Marking.
8822028	- Carton, Packing, Ammunition, for Case Section, Ballistic, Nose.
8821405	- Fuze T2078.
8825073	- Case Section, Ballistic, Nose.
8825110	- Skin, Nose, Tip.
8825158	- Bolt Assembly, Nose Tip.
EL-8842563	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The Case Section Ballistic Nose shall comply with all requirements specified on the drawing (dwg) 8825073, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Mating assembly. The assembly shall be capable of mating mechanically and electrically with the assembly, Fuze, T2078 (dwg 8821405) when tested as specified in 4.4.1.

3.6 Electrical resistance. The resistance of each circuit shall not exceed the maximum value listed in Table I when tested as specified in 4.4.2.

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Table I

Resistance Between pin and pin	Resistance (ohms)
P1-A to P2-A	0.040 maximum
P1-B to P1-C	0.040 maximum
P1-D to P2-D	0.040 maximum
P1-E to P2-E	0.040 maximum
P1-F to P1-G	0.040 maximum
P2-B to P2-C	0.040 maximum
P2-F to P2-G	0.040 maximum
P2-F to P1-G	0.040 maximum
P2-F to P1-E	0.040 maximum
P1-D to P1-C	0.040 maximum
P2-B to P1-C	0.040 maximum
P2-F to ground (external skin)	

3.7 Insulation resistance. Insulation resistance shall be 100 megohms minimum when tested at 500 volts ± 50 volts direct current (vdc) between any two isolated circuits, between isolated circuits and unused pins, and between the isolated circuits and the skin of the case section as specified in Table II. After exposure to the environmental conditions, as specified in 3.8.1, 3.8.2, 3.8.3, 3.8.4, and 3.8.5, the minimum insulation resistance shall be 20 megohms between the points specified in Table II. Testing shall be as specified in 4.4.3.

Table II

Resistance Between pin and pin	Insulation Resistance @ 500 ± 50 vdc
P1-A to P1-B, P1-G unused pins	20 megohms minimum
P1-B to P1-G, unused pins	20 megohms minimum
P1-G to unused pins	20 megohms minimum

Note: Unused pins are P1-H, P1-J, P1-K, P2-H, P2-J, and P2-K. Unused pins can be connected and checked as one pin.

3.8 Environmental testing. The assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The assembly shall also comply with 3.6 and 3.7 before and after each environmental test. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

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3.8.1 Waterproofness. The assembly when properly mated with T-2078 Fuze (dwg 8821405) shall now allow visible leakage inside the assembly when subjected to a water spray simulating heavy rain for a period of 15 minutes. After removal from the water spray, the assembly shall comply with 3.6, 3.7 and 3.9. Testing shall be as specified in 4.4.4.1.

3.8.2 Temperature shock. The assembly when properly mated with T-2078 Fuze (dwg 8821405) shall withstand, without damage, temperature shock between minus (-) 65 degrees Fahrenheit (°F) plus or minus (±) 5°F and plus (+) 160°F ± 5°F when tested as specified in 4.4.4.2. Upon completion of this test, the assembly shall comply with 3.6, 3.7 and 3.9.

3.8.3 Humidity. The assembly when properly mated with T-2078 Fuze (dwg 8821405) shall withstand exposure to a relative humidity of 95 + 5 percent at temperatures cycling from 80 ± 20°F to 160 ± 5°F to 80 ± 20°F without evidence of damage or deterioration when tested as specified in 4.4.4.3. Upon completion of this test, the assembly shall comply with 3.6, 3.7 and 3.9.

3.8.4 Corrosion resistance. The assembly when properly mated with T-2078 Fuze (dwg 8821405) shall withstand, without damage, exposure to salt spray conditions specified in 4.4.4.4. After salt spray, the assembly shall comply with 3.6, 3.7 and 3.9.

3.8.5 Low temperature cycling. The assembly when properly mated with T-2078 Fuze (dwg 8821405) shall withstand, without damage, temperature cycling of 80 ± 20°F to -65 ± 5°F to 80 ± 5°F when tested as specified in 4.4.4.5. After temperature cycling, the assembly shall comply with 3.6, 3.7 and 3.9.

3.8.6 Vibration. The assembly when properly mated with T-2078 Fuze (dwg 8821405) shall withstand without damage, vibration in each of three mutually perpendicular axes at the frequencies, amplitudes, temperatures*, and times specified in Table III. The assembly shall show no indication of continuity between pins P1-B and P2-F during vibration at each condition and shall comply with 3.6, 3.7 and 3.9 after being vibrated. Testing shall be as specified in 4.4.4.6.

Table III

Frequency Range Hertz(Hz)	Double amplitude or peak G level(+10%)	Cycle time to cover frequency range min. (± 0.5 minutes)	Time per axis minimum minutes 80 ± 20°F	160 ± 5°F	-65 ± 5°F
10-20-10	± 2G	3	30	10	10
20-75-20	0.036 inch	3	15	10	10
75-1000-75	± 10G	3	15	10	10

*The assembly shall be stabilized (see 6.4) at the applicable temperature before testing.

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3.8.7 Shock. The assembly when properly mated with T-2078 Fuze (dwg 8821405) shall withstand, without damage, a shock of 75 ± 8 gravitational units (g's) peak amplitude in each direction of three mutually perpendicular axes. The test shall be performed with the assembly at the various temperatures specified in Table IV. The assembly shall comply with 3.6, 3.7 and 3.9 after being shocked. Testing shall be as specified in 4.4.4.7.

Table IV

Peak shock amplitude	Time duration half amplitude	Temperature
75 ± 8 g's	9 ± 3 Milliseconds	-65 ± 5°F
75 ± 8 g's	9 ± 3 Milliseconds	80 ± 20°F
75 ± 8 g's	9 ± 3 Milliseconds	160 ± 5°F

3.8.8 Acceleration. The assembly when properly mated with T-2078 Fuze (dwg 8821405) shall withstand, without damage, acceleration in the axial and lateral directions for the magnitude and time duration specified in Table V. The assembly shall show no evidence of continuity between P1-B and P2-F during testing at each condition. After completion of the acceleration runs the assembly shall comply with the requirements of 3.6, 3.7 and 3.9. Testing shall be as specified in 4.4.4.8.

Table V

Acceleration - G's	Minimum duration - seconds	Direction of load
50 ± 5 g's	10	Axial forward
10 ± 1 g's	10	Axial aft
10 ± 1 g's	10	Lateral

3.9 Deterioration. Any internal or external components which show evidence of deterioration or corrosion shall provide reason to consider the assembly as having failed to meet the test to which it was subjected.

3.10 Welding. Welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of fusion penetration and soundness of weld deposit representative of good welding practice. They shall be free of irregularities which indicate lack of good welding practice on the part of the operator. All welding fluxes, scale, weld spatter, guide, and basic solutions shall be removed prior to application of any finish coat.

3.10.1 Welding operators. All fusion welding shall be performed by operators who are currently certified as Class "B" welders in accordance with Specification MIL-T-5021.

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3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 5 complete assemblies (see dwg 8825073) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VI. Tests shall be conducted in order specified in Table VI.

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Table VI

First article inspection and tests

Inspection or Test	Units 1&2	Units 3,4&5	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	X	X		X
2. Electrical resistance (see 3.6)	X	X		X
3. Insulation resistance (see 3.7)	X	X		X
4. Mating assembly (see 3.5)	X	X		X
5. Waterproofness (see 3.8.1)	X	X		X
6. Vibration (see 3.8.6)	X	X	X	
7. Mechanical shock (see 3.8.7)	X	X	X	
8. Temperature shock (see 3.8.2)	X	X	X	
9. Humidity (see 3.8.3)	X		X	
10. Corrosion resistance (see 3.8.4)	X		X	
11. Low temperature cycling (see 3.8.5)	X			X
12. Acceleration (see 3.8.8)	X	X	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VII and Table VIII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VIII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Bolt assembly, nose tip (see dwg 8825158 covering a detail of dwg 8825073).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper perpendicularity and concentricity of head to shaft	Gage
	102. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.2 Skin, nose tip (see dwg 8825110 covering a detail of dwg 8825073).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of datum diameter at large end	Gage
	102. Location of datum diameter at small end	Gage
	103. Improper welding (see 3.10)	Visual

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	104. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.3 Case section, ballistic, nose (see dwg 8825073).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Mismatch at joints greater than 0.030 inch step-up or 0.040 inch step-down	Gage
	2. Gap joints greater than 0.31 inch	Gage
Major:		
	101. Improper metal finish (interior and exterior)	Visual
	102. Evidence of poor workmanship (see 3.11)	Visual
	103. Improper paint coverage (when applicable)	Visual
	104. Improper paint color (when applicable)	Visual
	105. Voids, scratches, dents or protrusions showing parent metal (when applicable)	Visual
	106. Any part missing, loose or damaged	Visual/Manual
	107. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Dents- 0.010 inch maximum may be either concave or convex, maximum diameter with a minimum spacing of 6 inch edge to edge	Gage
	202. Scratches- 0.020 inch maximum depth, no limitation of width, maximum length of 10 inches	Gage
	203. Marking missing, mislocated or incorrect	Gage

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4.3.2.1.4 Carton, packing, ammunition, for case section, ballistic, nose, (see dwg 8822028).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Wrong sized bag	Visual
	102. Bag not sealed	Visual
	103. Air in bag not minimized before sealing	Visual
	104. Incorrect or insufficient cushioning material	Visual
	105. Evidence of poor workmanship (see 3.11)	Visual
	106. Marking missing or incorrect	Visual
	107. Packing material visibly wet	Visual
	108. Connector not wrapped correctly	Visual
	109. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.5 Container, packing, ammunition for domestic (CONUS) shipments of special weapons items, packing and marking (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect or insufficient cushioning material	Visual
	102. Marking improper, illegible, smeared, missing, or in wrong location	Visual
	103. Evidence of poor workmanship (see 3.11)	Visual
	104. Packing material visibly wet	Visual
Minor:	None defined.	

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Electrical resistance	Major (AQL 0.10%)
b. Insulation resistance	Major (AQL 0.10%)
c. Mating assembly	Major (AQL 0.10%)
d. Vibration	Major (AQL 0.10%)

Of these tests, d is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Electrical resistance, insulation resistance, and mating assembly. Each assembly in the lot shall be tested in accordance with 4.4.2, 4.4.3 and 4.4.1. If any item fails to comply with 3.6, 3.7 and 3.5, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration. A sample shall be selected at random in accordance with Table IX. The sample shall be tested in accordance with 4.4.4.8. If any item fails to comply with 3.8.8, the lot shall be rejected.

Table IX

Sample sizes for destructive testing

Number Required to be Delivered	Sample Size for No Failures
2-180	2
181-500	3
501-800	5
801-1300	7

Lot Size Required = Number Required to be Delivered + Sample Size.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8842563). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Mating assembly. The assembly shall be inspected on suitable inspection equipment when mounted to a fixture equivalent to the mating surface (maximum metal condition) on Fuze, T2078, drawing 8821405.

4.4.2 Electrical resistance. The electrical resistance of the specified circuits shall be determined by the use of a suitable electrical resistance measuring device.

4.4.3 Insulation resistance. The insulation resistance shall be determined by the use of an insulation resistance tester, capable of supplying the required voltage and with which a resistance in megohms can be obtained. The current shall be limited to a maximum of 5 milliamperes to protect human life.

4.4.4 Environmental testing (see 3.8).

4.4.4.1 Waterproofness. The assembly mated to T-2078 Fuze and mounted to simulate the installed conditions, shall be positioned horizontally, and impacted with a water spray directed within 15° of vertical. The rain test chamber temperature shall be maintained at 60 ± 20°F throughout the test period. A simulated rain fall of 10 ± 2 inches per hour shall be produced by means of a water spray nozzle. The rain fall shall be dispersed uniformly over the test area. During the test period the assembly shall be rotated once about the longitudinal axis. Following this test, the assembly shall be tested as specified in 4.4.2 and 4.4.3.

4.4.4.2 Temperature shock. The assembly, mounted to simulate the installed conditions, shall be placed within a test chamber where the temperature is -65 ± 5°F. The assembly shall be subjected to this temperature for a period of 4 hours minimum at the conclusion of which, and within 5 minutes, the assembly shall be transferred to a chamber having an internal temperature of 160 ± 5°F. The item shall be subjected to this temperature for a period of 4 hours minimum. This completes one cycle. The number of complete cycles shall be three, with a maximum of 5 minutes transfer time between test chambers at the conclusion of each cycle. At the conclusion of the third cycle the item shall be removed from the test chamber. Following this test the assembly shall be tested as specified in 4.4.2 and 4.4.3.

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4.4.4.3 Humidity. The assembly shall be mounted to simulate the installed conditions and shall be placed in the test chamber. The chamber temperature shall be 80 ± 20°F. The temperature shall be gradually raised to 160 ± 5°F during the first 2 hour period. The temperature of 160 ± 5°F shall be maintained during the next 6-hour period. During the following 16-hour period, the temperature in the chamber shall be gradually reduced to 80 ± 20°F. This completes one cycle. The relative humidity throughout the cycle shall be 95 ± 5 percent. The tests shall consist of 15 such cycles, (360 hours). The velocity of the air throughout the test chamber shall not exceed 150 feet per minute. Following this test, the assembly shall be tested as specified in 4.4.2 and 4.4.3.

4.4.4.4 Corrosion resistance. The assembly shall be mounted to simulate the installed conditions and shall then be subjected to the salt spray test as specified in MIL-E-5272, section 4.6, procedure I, for an exposure time of 50 hours. At the end of this time, the assembly shall be removed from the test chamber. Following this test, the assembly shall be tested as specified in 4.4.2 and 4.4.3.

4.4.4.5 Low temperature cycle. The assembly shall be mounted to simulate installed conditions and shall then be placed in the test chamber. The chamber temperature shall be 80 ± 20°F. During the first 2-hour period the temperature shall be gradually reduced to -65 ± 5°F. The temperature of -65 ± 5°F shall be maintained for 4 hours. During the following 2-hour period, the temperature shall be gradually increased to 80 ± 20°F. The test chamber shall be maintained in this condition for 4 hours. This completes one cycle. The test assembly shall be subjected to 10 cycles. Following this test, the assembly shall be tested as specified in 4.4.2 and 4.4.3.

4.4.4.6 Vibration. The assembly shall be mounted to simulate installed conditions and shall be rigidly mounted to a fixture capable of supporting the assembly in each of three mutually perpendicular axes, with the fixture mounted on a suitable vibration producing apparatus. The assembly shall be vibrated at the frequencies, amplitudes, temperatures, and for the time specified in Table III in each direction of three mutually perpendicular axes. The circuit between the designated terminals shall be monitored during the vibration using instrumentation capable of detecting closures lasting 20 microseconds. Following this test, the assembly shall be tested as specified in 4.4.2 and 4.4.3.

4.4.4.7 Shock. The assembly mounted to simulate installed conditions shall be mounted to a fixture capable of supporting the assembly in each of three mutually perpendicular axes. The fixture shall be mounted to a suitable shock produced device. The assembly shall be shock-tested three

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times as specified in Table IV in each direction of three mutually perpendicular axes. Following this test, the assembly shall be tested as specified in 4.4.2 and 4.4.3.

4.4.4.8 Acceleration. The assembly mounted to simulate installed conditions, shall be rigidly mounted to a fixture capable of supporting the assembly in the axial and lateral directions with the fixture mounted on a suitable acceleration producing device. The assembly shall be subjected to acceleration for the times and in the directions specified in Table V. The circuit between the designated terminals shall be monitored during the vibration using instrumentation capable of detecting closures lasting 20 microseconds. Following this test, the assembly shall be tested as specified in 4.4.2 and 4.4.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The assembly shall be packaged in accordance with dwg 8822028 using the specified moisture barrier.

5.1.2 Level C. The assembly shall be packaged in accordance with dwg 8822028 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The assembly shall be packed in accordance with dwg 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwg 8822028 and 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

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6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1682 dated 18 December 1961.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1115-A052

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